

L1: Introduction to MySQL

MySQL



- MySQL ("My S-Q-L") is an open-source relational database management system (RDBMS).
- Its name is a combination of "My", the name of co-founder Michael Widenius's daughter, and "SQL", the abbreviation for Structured Query Language.
- <https://www.mysql.com/>

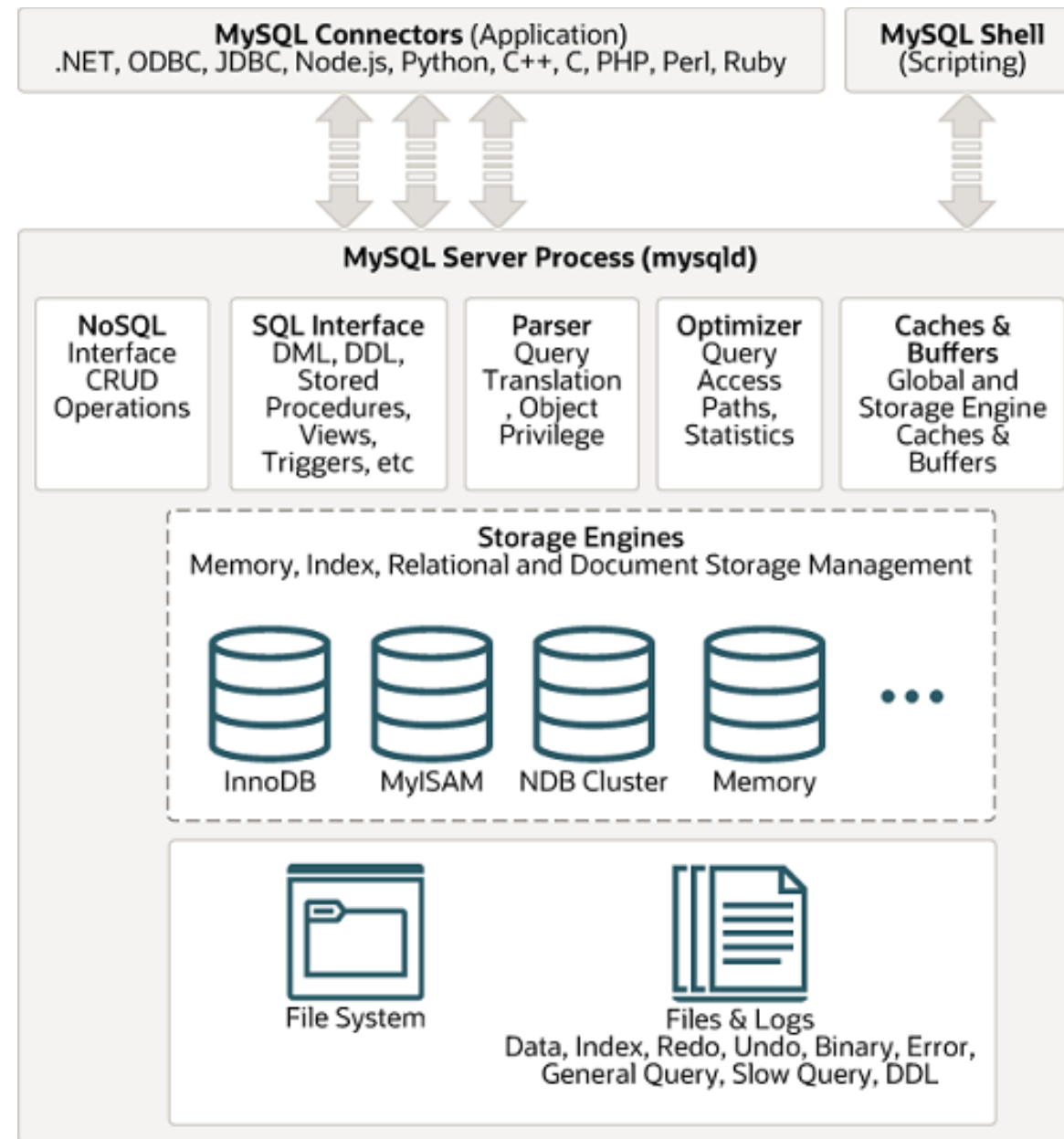
MySQL Architecture

Client

- The client is the application that interacts with the MySQL database server.
- The client can be a standalone application, a web application, or any program that needs a database.
- The client sends SQL queries to the MySQL server for processing.

Server

- The server is the MySQL database management system responsible for storing, managing, and processing data.
- The server receives SQL queries, processes them, and returns the result sets.
- The server manages multiple clients' data storage, security, and concurrent access.

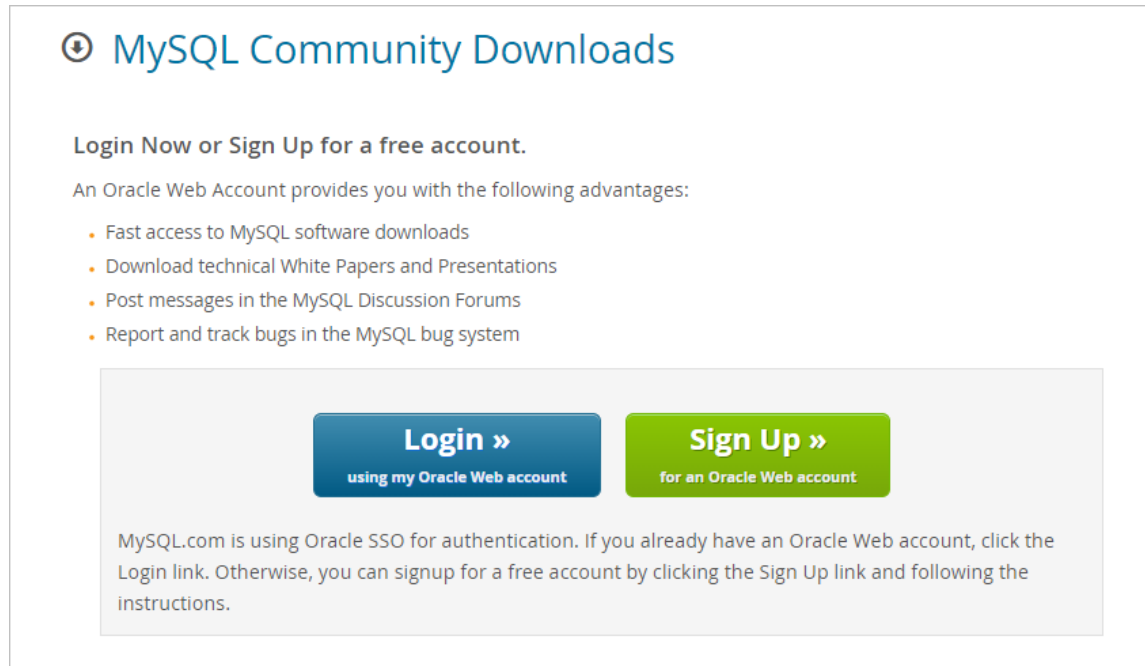


Installation

- **MySQL Server**: MySQL Community Server 8.0.40
<https://dev.mysql.com/downloads/mysql/>
- **MySQL Workbench** 8.0.40
<https://dev.mysql.com/downloads/workbench/>
- Note: You need "**Server**" before "**Workbench**"

Register Oracle Account (Optional)

- Register for an Oracle Account (using your MU email)
 - You can use your MU student emails
 - Use Faculty of ICT as your workplace contact
- Verify your email
- Download and install



The screenshot shows the 'MySQL Community Downloads' page. At the top, there's a header with a download icon and the text 'MySQL Community Downloads'. Below this, a prompt says 'Login Now or Sign Up for a free account.' followed by a paragraph: 'An Oracle Web Account provides you with the following advantages:'. A bulleted list follows, detailing benefits like fast access to MySQL software, downloading technical documents, posting in forums, and reporting bugs. At the bottom, there are two prominent buttons: a blue 'Login »' button with the subtext 'using my Oracle Web account' and a green 'Sign Up »' button with the subtext 'for an Oracle Web account'. A final paragraph at the very bottom explains that MySQL.com uses Oracle SSO for authentication and provides instructions on how to proceed based on whether the user already has an Oracle Web account.

MySQL Community Downloads

Login Now or Sign Up for a free account.

An Oracle Web Account provides you with the following advantages:

- Fast access to MySQL software downloads
- Download technical White Papers and Presentations
- Post messages in the MySQL Discussion Forums
- Report and track bugs in the MySQL bug system

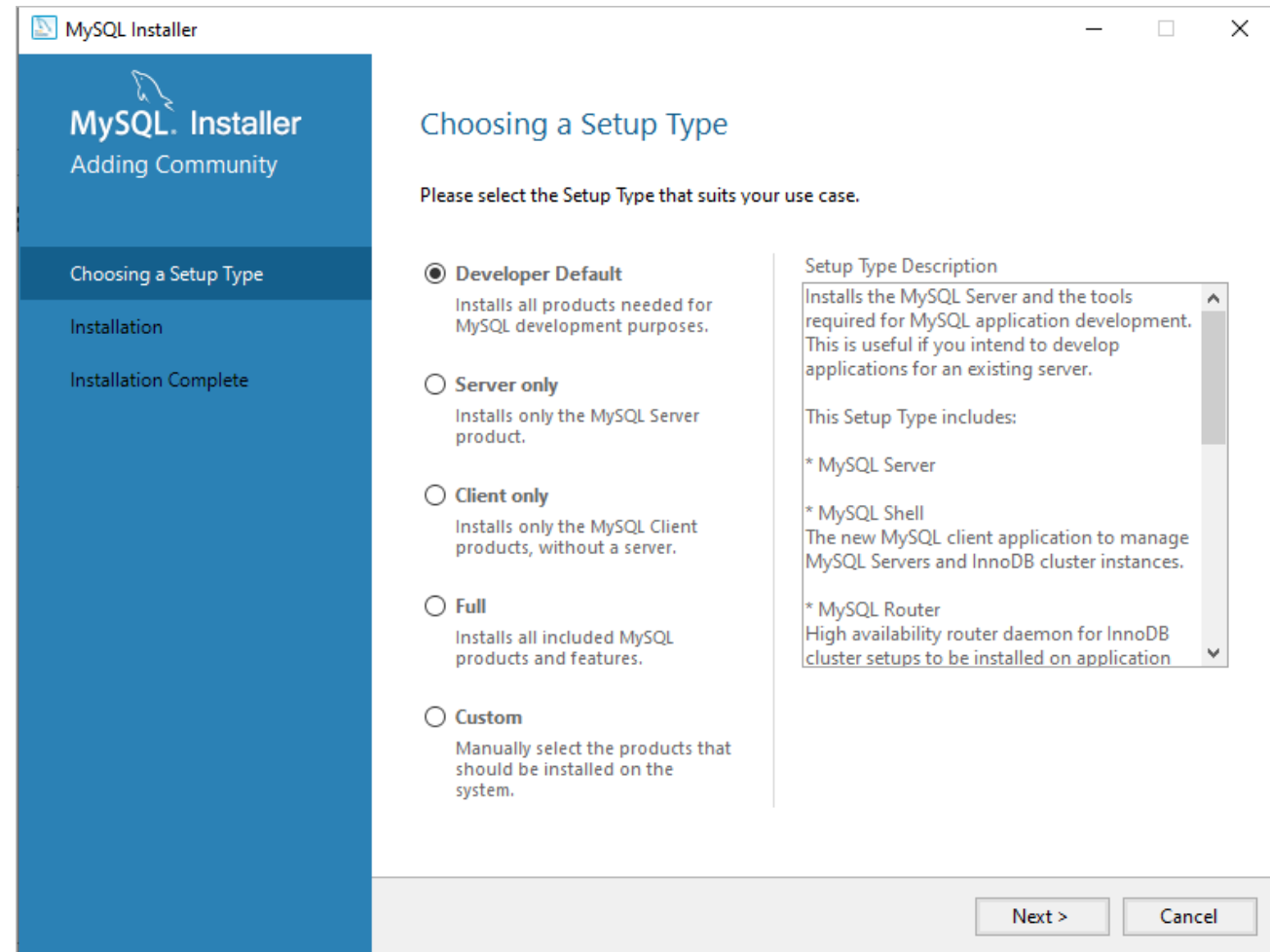
Login »
using my Oracle Web account

Sign Up »
for an Oracle Web account

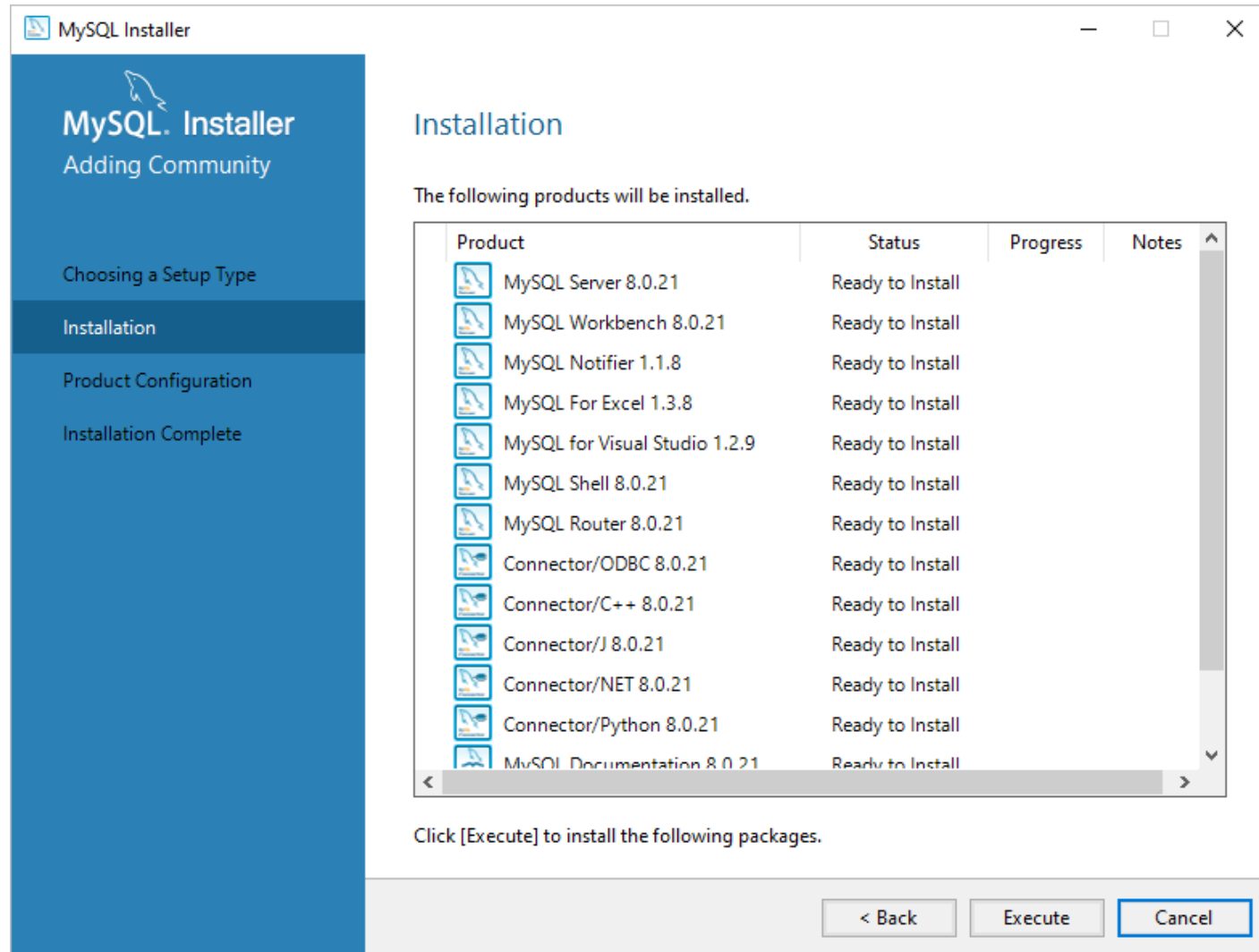
MySQL.com is using Oracle SSO for authentication. If you already have an Oracle Web account, click the Login link. Otherwise, you can signup for a free account by clicking the Sign Up link and following the instructions.

MySQL - Developer Default

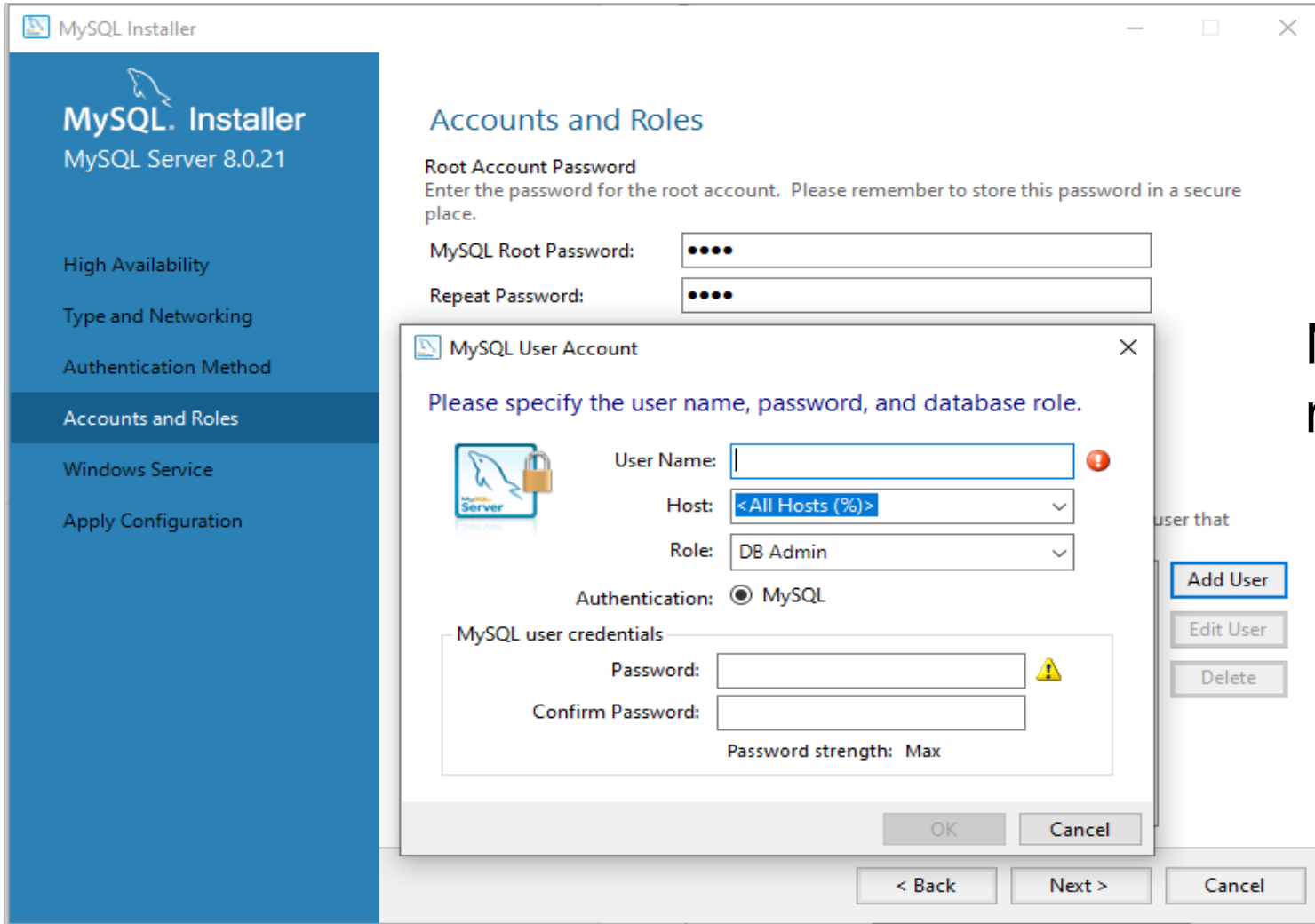
Note: Your computer may need to install extra packages required by the MySQL workbench



MySQL – Installation



MySQL – Set Up Root User



The screenshot shows the MySQL Installer window for MySQL Server 8.0.21. The left sidebar lists the installation steps: High Availability, Type and Networking, Authentication Method, Accounts and Roles (selected), Windows Service, and Apply Configuration. The main area is titled 'Accounts and Roles' and contains the 'Root Account Password' section with two password input fields. Overlaid on this is a 'MySQL User Account' dialog box. This dialog prompts the user to specify the user name, password, and database role. It includes fields for 'User Name', 'Host' (set to '<All Hosts (%)>'), 'Role' (set to 'DB Admin'), and 'Authentication' (set to 'MySQL'). There is also a section for 'MySQL user credentials' with 'Password' and 'Confirm Password' fields, and a 'Password strength' indicator set to 'Max'. Buttons for 'Add User', 'Edit User', and 'Delete' are visible on the right side of the dialog, along with 'OK' and 'Cancel' buttons at the bottom.

MySQL Installer
MySQL Server 8.0.21

High Availability
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Apply Configuration

Accounts and Roles

Root Account Password
Enter the password for the root account. Please remember to store this password in a secure place.

MySQL Root Password:

Repeat Password:

MySQL User Account

Please specify the user name, password, and database role.

User Name:

Host:

Role:

Authentication: ☒ MySQL

MySQL user credentials

Password:

Confirm Password:

Password strength: Max

Add User
Edit User
Delete

OK Cancel

< Back Next > Cancel

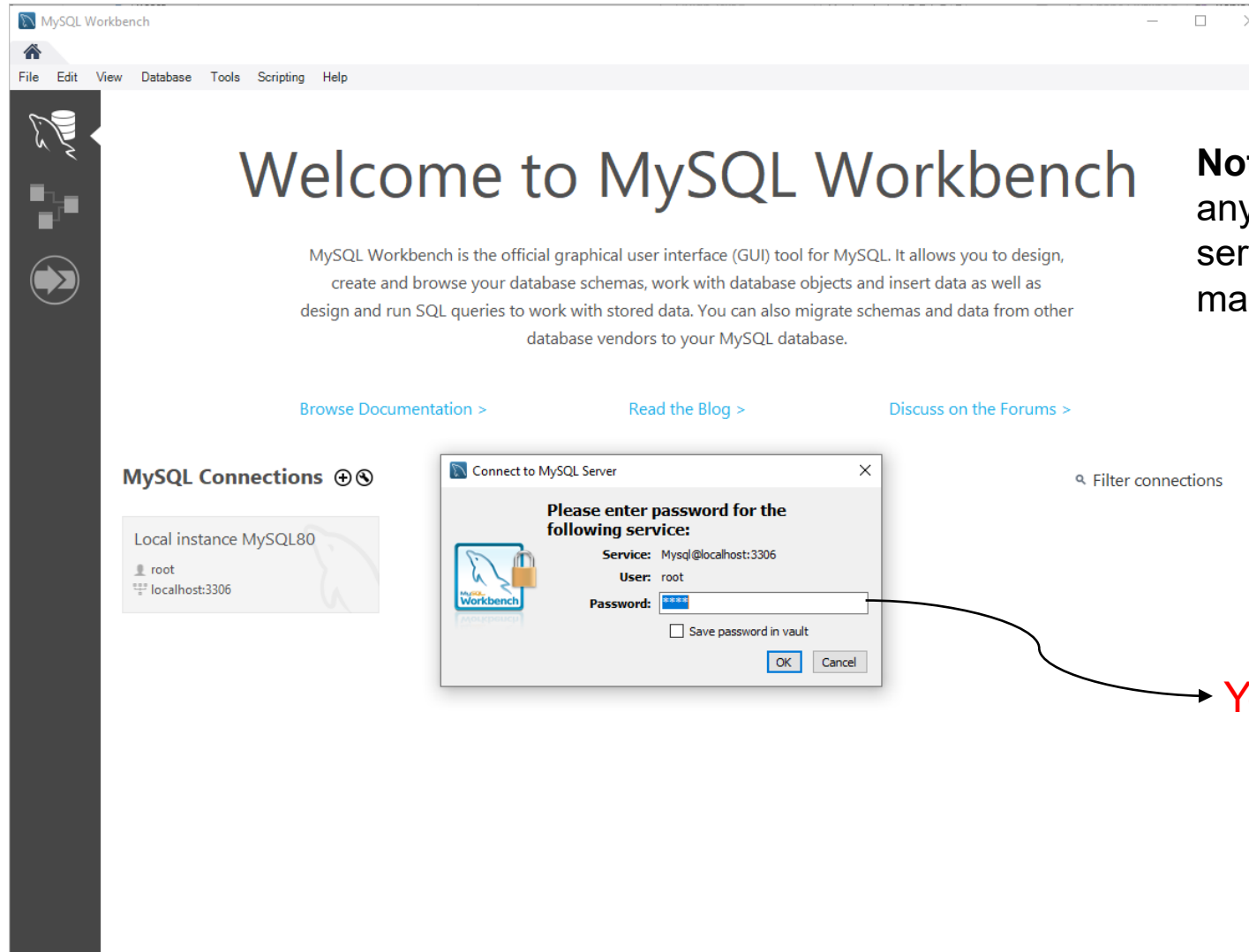
Note: Make sure you **remember** root and password

PW: **ict555**

For more details

- You can follow the following videos:
 - Windows OS (Link: https://youtu.be/OM4aZJW_Ojs)
 - macOS (Link: <https://youtu.be/-BDbOOY9jsc>)

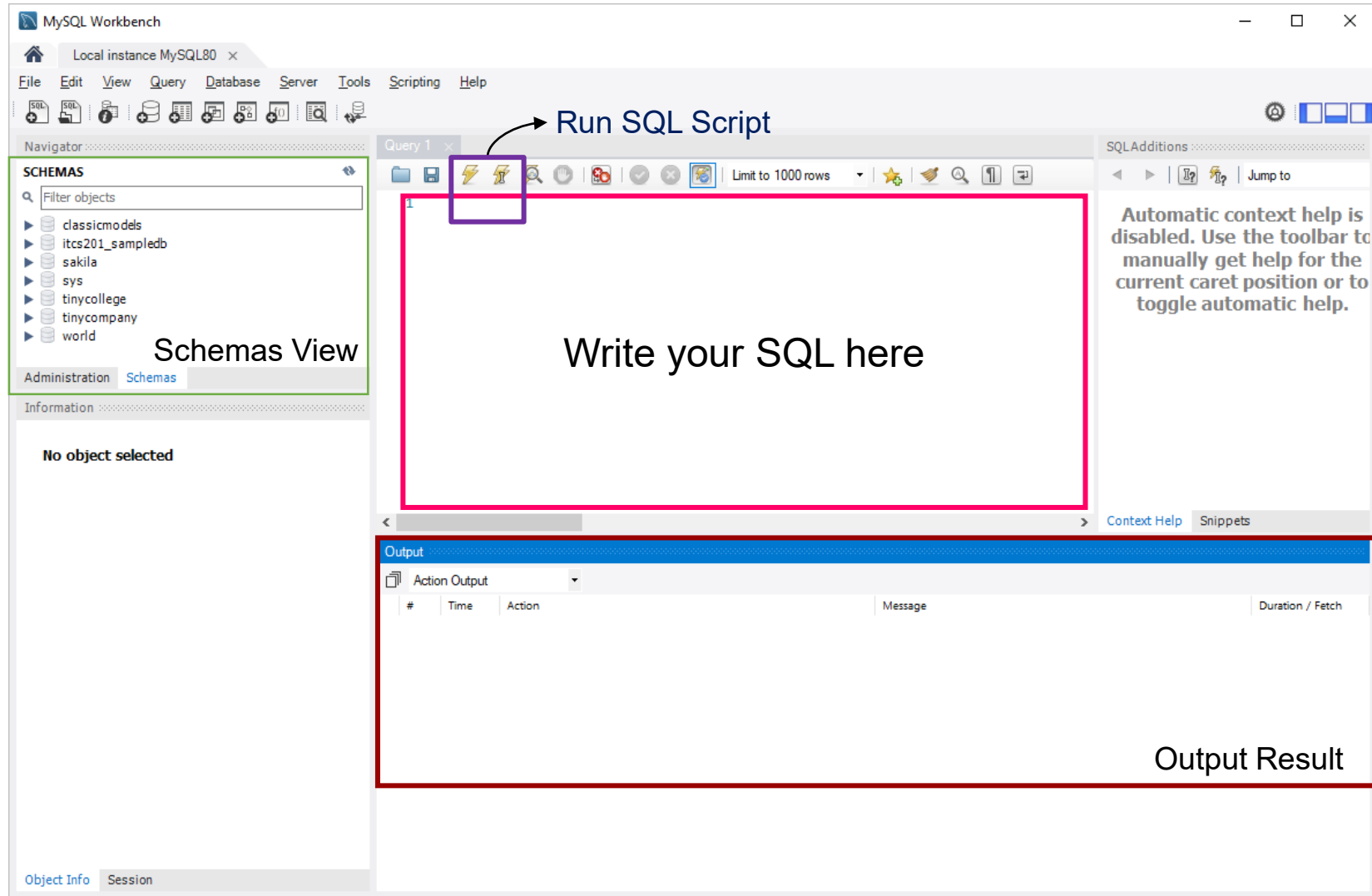
Start Your MySQL



Note: If you open Workbench and there is no any “Local Instance” found, make sure MySQL server is installed correctly or you may need to manually start MySQL server [\[ref\]](#)

Your root password

MySQL Workbench



Comment

Syntax 1

```
-- This is a single-line comment
```

Syntax 2

```
# This is another single-line comment
```

Syntax 3

```
/* This is a  
multi-line comment */
```

L1: Column & Constraints

1.1 Alter Columns in Existing Table

- To Add / Modify / Rename / Drop a column
- The command that we will use is `ALTER`.
- See more: <https://www.mysqltutorial.org/mysql-basics/mysql-alter-table/>

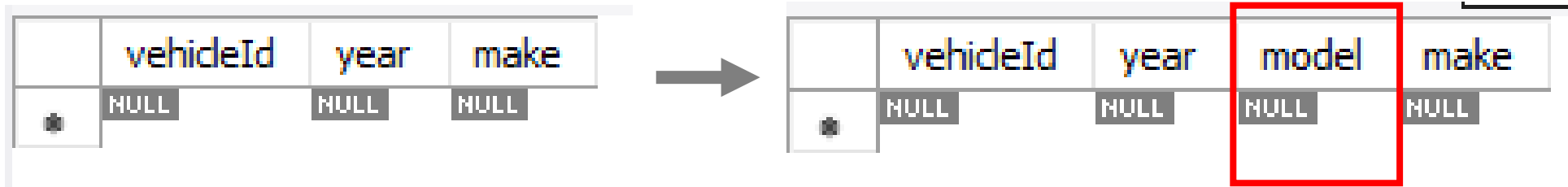
Alter : Add

Syntax

```
ALTER TABLE table_name  
ADD new_column_name column_definition  
[FIRST | AFTER column_name]
```

Example

```
ALTER TABLE vehicles  
ADD model VARCHAR(100) NOT NULL  
AFTER year;
```



Alter : Modify

Syntax

```
ALTER TABLE table_name  
MODIFY column_name column_definition  
[FIRST | AFTER column_name]
```

Example

```
ALTER TABLE vehicles  
MODIFY model VARCHAR(20) NOT NULL  
AFTER vehicleId;
```

Field	Type	Null	Key	Default	Extra
vehicleId	int	NO	PRI	NULL	
year	int	NO		NULL	
model	varchar(100)	NO		NULL	
make	varchar(100)	NO		NULL	



Field	Type	Null	Key	Default	Extra
vehicleId	int	NO	PRI	NULL	
model	varchar(20)	NO		NULL	
year	int	NO		NULL	
make	varchar(100)	NO		NULL	

Alter : Rename (Column)

Syntax

```
ALTER TABLE table_name  
CHANGE COLUMN original_name new_name column_definition  
[FIRST | AFTER column_name]
```

Example

```
ALTER TABLE vehicles  
CHANGE COLUMN model model_ID VARCHAR(10) NOT NULL;
```

Field	Type	Null	Key	Default	Extra
vehicleId	int	NO	PRI	NULL	
model	varchar(20)	NO		NULL	
year	int	NO		NULL	
make	varchar(100)	NO		NULL	



Field	Type	Null	Key	Default	Extra
vehicleId	int	NO	PRI	NULL	
model_ID	varchar(10)	NO		NULL	
year	int	NO		NULL	
make	varchar(100)	NO		NULL	

Alter : Rename (Table)

Syntax

```
ALTER TABLE table_name  
RENAME TO new_table_name;
```

Example

```
ALTER TABLE vehicles  
RENAME TO cars;
```

Alter : Drop

Syntax

```
ALTER TABLE table_name  
DROP COLUMN column_name
```

Example

```
ALTER TABLE vehicles  
DROP COLUMN model_id;
```

Field	Type	Null	Key	Default	Extra
vehideId	int	NO	PRI	NULL	
model_ID	varchar(10)	NO		NULL	
year	int	NO		NULL	
make	varchar(100)	NO		NULL	



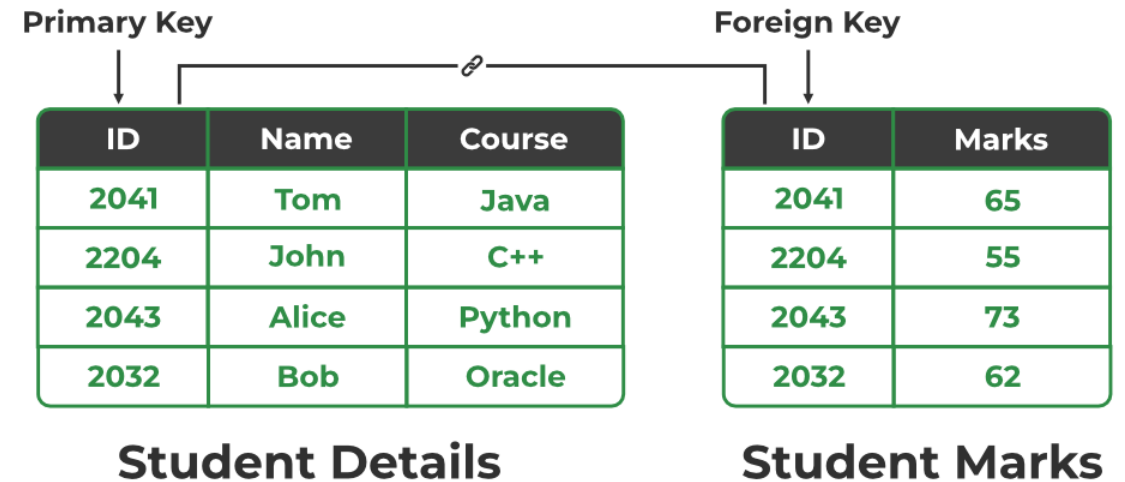
Field	Type	Null	Key	Default	Extra
vehideId	int	NO	PRI	NULL	
year	int	NO		NULL	
make	varchar(100)	NO		NULL	

MySQL DATA TYPES

DATE TYPE	SPEC	DATA TYPE	SPEC
CHAR	String (0 - 255)	INT	Integer (-2147483648 to 2147483647)
VARCHAR	String (0 - 255)	BIGINT	Integer (-9223372036854775808 to 9223372036854775807)
TINYTEXT	String (0 - 255)	FLOAT	Decimal (precise to 23 digits)
TEXT	String (0 - 65535)	DOUBLE	Decimal (24 to 53 digits)
BLOB	String (0 - 65535)	DECIMAL	"DOUBLE" stored as string
MEDIUMTEXT	String (0 - 16777215)	DATE	YYYY-MM-DD
MEDIUMBLOB	String (0 - 16777215)	DATETIME	YYYY-MM-DD HH:MM:SS
LONGTEXT	String (0 - 4294967295)	TIMESTAMP	YYYYMMDDHHMMSS
LOBLOB	String (0 - 4294967295)	TIME	HH:MM:SS
TINYINT	Integer (-128 to 127)	ENUM	One of preset options
SMALLINT	Integer (-32768 to 32767)	SET	Selection of preset options
MEDIUMINT	Integer (-8388608 to 8388607)	BOOLEAN	TINYINT(1)

1.2 Primary Key & Foreign Key

Features	Primary Keys	Foreign Keys
In tables	Part of a parent table	Part of a child table, always links back to a primary key in a parent table
Number	Only one per parent table	Tables can have multiple foreign keys
Values	Must have a value, and every item must be identified	Can have the value of NULL
Ease	Cannot be removed from the table	Can be removed from the table



- Primary key is a column or a set of columns that **uniquely** identifies each row in the table.
- A foreign key is one or more columns in a table that **references** the primary key of another, creating a link between them.

Assign Primary key (with create table)

Syntax 1

```
CREATE TABLE table_name(  
    column1 datatype PRIMARY KEY,  
    column2 datatype,  
    ...  
);
```

Syntax 2

```
CREATE TABLE table_name(  
    column1 datatype,  
    column2 datatype,  
    .../  
    PRIMARY KEY(column1)  
);
```

Assign Primary key (with create table)

Example

```
CREATE TABLE products(  
    id INT PRIMARY KEY,  
    name VARCHAR(255) NOT NULL  
);
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	
name	varchar(255)	NO		NULL	

Assign Primary key (alter table)

Syntax

```
ALTER TABLE table_name  
ADD PRIMARY KEY(column_name);
```

Example

```
ALTER TABLE products2  
ADD PRIMARY KEY(id);
```

Field	Type	Null	Key	Default	Extra
id	int	YES		NULL	
name	varchar(255)	NO		NULL	



Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	
name	varchar(255)	NO		NULL	

Remove Primary key

Syntax

```
ALTER TABLE table_name  
DROP PRIMARY KEY;
```

Example

```
ALTER TABLE products2  
DROP PRIMARY KEY;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	
name	varchar(255)	NO		NULL	



Field	Type	Null	Key	Default	Extra
id	int	YES		NULL	
name	varchar(255)	NO		NULL	

Assign Foreign key

Syntax: A foreign key constraint in the CREATE TABLE or ALTER TABLE statement.

```
FOREIGN KEY (column_name, ...)
REFERENCES parent_table(column_name, ...)
[ON DELETE reference_option]
[ON UPDATE reference_option]
```

Assign Primary key (with create table)

Example

```
-- Create a 'Student' table with a Primary Key
CREATE TABLE Student (
  StudentID INT NOT NULL,
  FirstName VARCHAR(50) NOT NULL,
  LastName VARCHAR(50) NOT NULL,
  PRIMARY KEY (StudentID)
);

-- Create an 'Enrollment' table with a Foreign Key
CREATE TABLE Enrollment (
  EnrollmentID INT NOT NULL,
  StudentID INT NOT NULL,
  CourseName VARCHAR(100) NOT NULL,
  EnrollmentDate DATE NOT NULL,
  FOREIGN KEY (StudentID) REFERENCES Student (StudentID)
);
```

Assign Primary key (with create table)

Example's result

Student Table

Field	Type	Null	Key	Default	Extra
StudentID	int	NO	PRI	NULL	
FirstName	varchar(50)	NO		NULL	
LastName	varchar(50)	NO		NULL	

Enrollment Table

Field	Type	Null	Key	Default	Extra
EnrollmentID	int	NO		NULL	
StudentID	int	NO	MUL	NULL	
CourseName	varchar(100)	NO		NULL	
EnrollmentDate	date	NO		NULL	

Assign Foreign key (reference_option)

reference_option	Description
CASCADE	Delete or update the row from the parent table and automatically delete or update the matching rows in the child table. Both ON DELETE CASCADE and ON UPDATE CASCADE are supported.
SET NULL	Delete or update the row from the parent table and set the foreign key column or columns in the child table to NULL. Both ON DELETE SET NULL and ON UPDATE SET NULL clauses are supported.
RESTRICT	Rejects the delete or update operation for the parent table. Specifying RESTRICT (or NO ACTION) is the same as omitting the ON DELETE or ON UPDATE clause.
NO ACTION	The MySQL Server rejects the delete or update operation for the parent table if there is a related foreign key value in the referenced table. In MySQL, foreign key constraints are checked immediately, so NO ACTION is the same as RESTRICT.
SET DEFAULT	This action is recognized by the MySQL parser, but both InnoDB and NDB reject table definitions containing ON DELETE SET DEFAULT or ON UPDATE SET DEFAULT clauses.

1.3 Constraints

What are Constraints?

- Rules for the data in a table.
- The action is terminated, if there is a violation of any constraints.
- They can be specified when CREATE TABLE or ALTER TABLE.

We can apply other constraints to the column by the same method as the primary key.

If you define a UNIQUE constraint without specifying a name, MySQL automatically generates a name for it.

Constraints (Types)

CONSTRAINT	DESCRIPTION
NOT NULL	In MySQL NOT NULL constraint allows to specify that a column can not contain any NULL value. MySQL NOT NULL can be used to CREATE and ALTER a table.
UNIQUE	The UNIQUE constraint in MySQL does not allow to insert a duplicate value in a column. The UNIQUE constraint maintains the uniqueness of a column in a table. More than one UNIQUE column can be used in a table.
PRIMARY KEY	A PRIMARY KEY constraint for a table enforces the table to accept unique data for a specific column and this constraint creates a unique index for accessing the table faster.
FOREIGN KEY	A FOREIGN KEY in MySQL creates a link between two tables by one specific column of both tables. The specified column in one table must be a PRIMARY KEY and referred by the column of another table known as FOREIGN KEY.
CHECK	A CHECK constraint controls the values in the associated column. The CHECK constraint determines whether the value is valid or not from a logical expression.
DEFAULT	In a MySQL table, each column must contain a value (including a NULL). While inserting data into a table, if no value is supplied to a column, then the column gets the value set as DEFAULT.

Assign Constraints (with CREATE TABLE)

Syntax

```
[CONSTRAINT constraint_name]  
<CONSTRAINT> (column_list, ...)
```

Example

```
CREATE TABLE suppliers (  
    supplier_id INT AUTO_INCREMENT,  
    name VARCHAR(255) NOT NULL,  
    phone VARCHAR(15) NOT NULL UNIQUE,  
    address VARCHAR(255) NOT NULL,  
    PRIMARY KEY (supplier_id),  
    CONSTRAINT uc_name_address UNIQUE (name, address)  
);
```


Assign Constraints (with ALTER)

Syntax

```
ALTER TABLE table_name  
ADD CONSTRAINT constraint_name  
<CONSTRAINT> (column_list);
```

Example

```
ALTER TABLE suppliers  
ADD CONSTRAINT uc_name_address  
UNIQUE (name,address);
```